

Upper-Crustal Seismic Anisotropy and Its Geological Controls in Southeastern Korea from Local Earthquake Data

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ABSTRACT

We investigated upper-crustal seismic anisotropy in the southeastern Korean Peninsula using local-earthquake shear-wave splitting (SWS) recorded by the Gyeongju Hi-density Broadband Seismic Network. A total of 4611 reliable SWS measurements were analyzed to characterize fast-polarization directions (ϕ) and delay times (δt) in a tectonically complex intraplate setting. Spatial averaging reveals a dominant northeast–southwest (NE–SW) fast orientation across much of the study area, with local rotations to north–south to north-northeast–south-southwest (NNE–SSW) around the Yangsan fault system and to east–west (E–W) near the Ulsan fault; east of the Ulsan fault, NE–SW and E–W orientations coexist. Delay-time tomography shows elevated δt values near the 2016 Gyeongju earthquake sequence and east of the Ulsan fault, where the high-anisotropy zone terminates along the Yeonil tectonic line. These patterns indicate that upper-crustal anisotropy is controlled by multiple sources. Localized high anisotropy near the Naenam fault suggests fault-related deformation associated with the Gyeongju sequence. East of the Ulsan fault, anisotropy likely reflects a combination of inherited Miocene basin-related structures and the present regional stress field. The widespread NE–SW component, therefore, is not attributed to a single origin but is likely to record domain-dependent contributions from inherited structures and, locally, shallow stress conditions distinct from the deeper east-northeast–west-southwest regional compression. These results provide a new baseline for understanding upper-crustal anisotropy in a diffuse intraplate setting.

KEY POINTS

- We address whether anisotropy mainly reflects inherited fabrics, stress, or both.
- The observed anisotropy records domain-dependent structural and stress contributions.
- Our results provide a new reference for upper-crustal anisotropy in intraplate settings.

Supplemental Material

INTRODUCTION

Upper-crustal anisotropy in intraplate fault zones may record both long-lived structural inheritance and ongoing deformation, but separating these signals remains difficult. The southeastern Korean Peninsula provides a useful setting for addressing this problem because major fault systems, contrasting tectonic domains, and recent moderate seismicity occur within a relatively compact region. The Gyeongju area is particularly suitable, as the 2016 M_w 5.5 (M_L 5.8) earthquake sequence and adjacent fault systems allow anisotropy to be examined across

structurally distinct domains. In this context, local-earthquake shear-wave splitting (SWS) offers an effective framework for evaluating the relative roles of inherited fabrics and the present-day stress field.

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SWS and crustal anisotropy

Seismic anisotropy provides important constraints on the physical state, deformation, and evolution of the Earth's crust (Crampin, 1994; Gao *et al.*, 1995; Okaya *et al.*, 2016; Gao *et al.*, 2019; Pastori *et al.*, 2019; Hu *et al.*, 2020). When shear waves propagate through an anisotropic medium, they split into fast and slow components, a phenomenon known as SWS (Kaneshima, 1990; Savage *et al.*, 2010; Giannopoulos *et al.*, 2012; Sharma *et al.*, 2017). SWS analysis yields two key parameters: the fast polarization direction (ϕ), which reflects the preferred orientation of anisotropic fabrics, and the delay time between the split shear waves (δt), which reflects the integrated strength of anisotropy along the ray path.

Crustal anisotropy commonly arises from two end-member controls: structural anisotropy and stress-induced anisotropy (Crampin, 2003; Boness and Zoback, 2006; Gao and Wu, 2008; Gao *et al.*, 2012). Structural anisotropy is associated with geological fabrics such as faults, lithologic boundaries, and long-lived deformation structures that persist through geologic time (Wu *et al.*, 2007). In contrast, stress-induced anisotropy reflects the present-day stress field, in which distributed microcracks tend to align in preferred orientations under differential stress (Crampin and Booth, 1989; Crampin and Zatsepin, 1995; Balfour *et al.*, 2005; Crampin and Gao, 2018; Cochran *et al.*, 2020). Distinguishing between these contributions is essential for interpreting fault-zone deformation and present-day crustal stress.

Seismotectonic setting of the southeastern Korean Peninsula

The study area is located in the southeastern Korean Peninsula (Fig. 1) within the Gyeongsang basin (GB), which consists mainly of nonmarine sedimentary rocks and volcanic units (Chough and Sohn, 2010; Kang and Paik, 2013). The region is cut by several major tectonic structures, including the Yangsan fault (YSF), the Ulsan fault (USF), and the Yeonil tectonic line (YTL) (Hwang *et al.*, 2007; Son *et al.*, 2015; Cheon *et al.*, 2019; Cheon, Kim, *et al.*, 2023; Cheon, Shin, *et al.*, 2023). The Yangsan fault system is composed of the YSF and its major subsidiary strands, the Miryang fault (MiRF), Moryang fault (MoRF), Dongrae fault (DRF), and Ilgwang fault (IGF). These north-northeast (NNE)-trending faults share similar orientations, but the YSF is the principal structure, extending for approximately 200 km and accommodating a cumulative displacement of ~ 21 km (Hwang *et al.*, 2004, 2007; Choi *et al.*, 2009; Cheon *et al.*, 2020).

The USF is an east-dipping reverse fault that connects with the YSF to form a λ -shaped structural pattern, where north-northwest (NNW)-striking USF joins the NNE-striking YSF (Park, 2004; Han *et al.*, 2009). East of the USF, Miocene deformation produced several basins and associated northeast-southwest (NE-SW)-trending basin-bounding faults, and the YTL forms an important structural boundary in this domain (Kim

et al., 1998; Son *et al.*, 2013; Son *et al.*, 2015; Cheon, Kim, *et al.*, 2023; Cheon, Shin, *et al.*, 2023). The present-day maximum horizontal compressive stress in southeastern Korea is generally east-northeast-west-southwest (ENE-WSW), although it locally rotates toward east-west (E-W) east of the USF (Chang *et al.*, 2010; Soh *et al.*, 2018; Seo, 2021). This tectonic setting suggests that both inherited structures and the present stress field may influence crustal anisotropy in the region.

Seismicity in the study area is concentrated in two main clusters (Fig. 2). One is the 2016 Gyeongju earthquake sequence, which occurred on the Naenam fault, a subsurface subsidiary fault of the YSF (Kim, Kang, *et al.*, 2016; Kim, Rhie, *et al.*, 2016; Kim *et al.*, 2018; Woo *et al.*, 2019; Lim *et al.*, 2021; Heo *et al.*, 2024). The other is a cluster of earthquakes in the hanging wall east of the USF. The Naenam fault strikes north-northeast-south-southwest (NNE-SSW) at depths of ~ 12 – 17 km, whereas the eastern-USF cluster mainly occurs at depths of ~ 8 – 15 km and delineates an ENE-dipping fault plane (Heo *et al.*, 2024).

Motivation and objectives of this study

Previous studies have shown that spatial and temporal variations in SWS can track fault evolution, stress perturbations, and earthquake-related deformation in a variety of tectonic environments (Gerst, 2003; Paulssen, 2004; Peng and Ben-Zion, 2004; Boness and Zoback, 2006; Hurd and Bohnhoff, 2012; Polat *et al.*, 2012; Eken *et al.*, 2013; Li *et al.*, 2014; Pastori *et al.*, 2019; Chacón-Hernández *et al.*, 2021; Kapetanidis *et al.*, 2021; Kaviris *et al.*, 2021; Kim and Kim, 2023; Pastori *et al.*, 2023). On the Korean Peninsula, SWS studies remain limited. At the regional scale, Kang and Shin (2009) presented the first systematic SWS analysis of southern Korea using ScS phases, providing a broader regional, deeper-scale context. By contrast, the controls on upper-crustal anisotropy associated with local seismicity and major fault systems in southeastern Korea remain poorly constrained. Disentangling these controls is particularly difficult in settings characterized by spatially diffuse intraplate seismicity, where short observational windows hinder the assessment of temporal changes.

To address this problem, the Gyeongju Hi-density Broadband Seismic Network (GHBSN) was deployed to resolve the 3D geometry and slip behavior of the causative fault (Heo *et al.*, 2024). Using GHBSN data, this study characterizes the spatial variation of upper-crustal seismic anisotropy in the southeastern Korean Peninsula and evaluates its relationship with major fault systems, recent seismicity, and the regional stress field. In particular, we assess whether the observed anisotropy is primarily controlled by inherited structural fabrics, present-day stress, or a combination of both.

DATA AND METHODS

Data overview

The GHBSN was established after the 2016 Gyeongju earthquake to improve seismic monitoring in southeastern Korea.

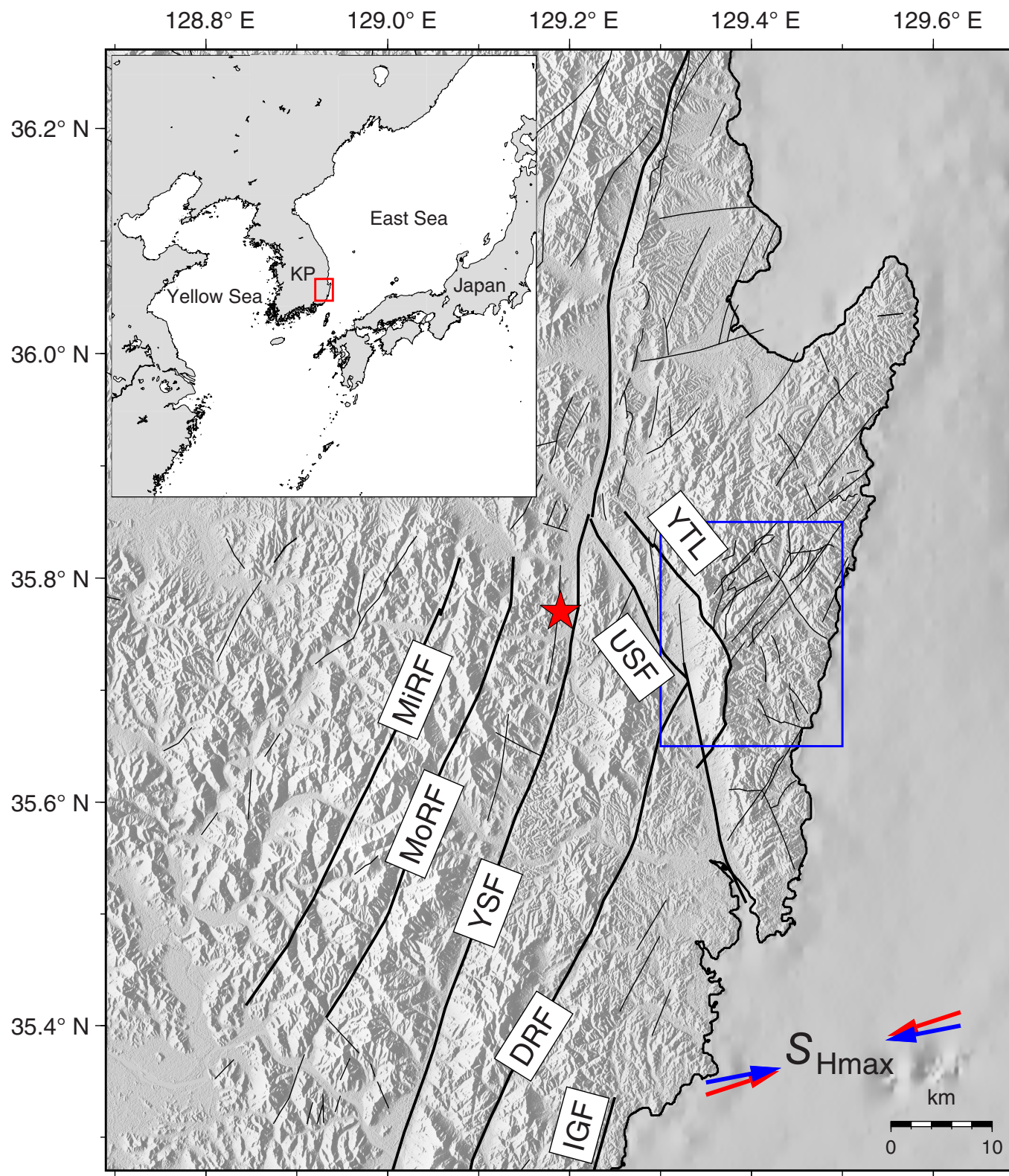
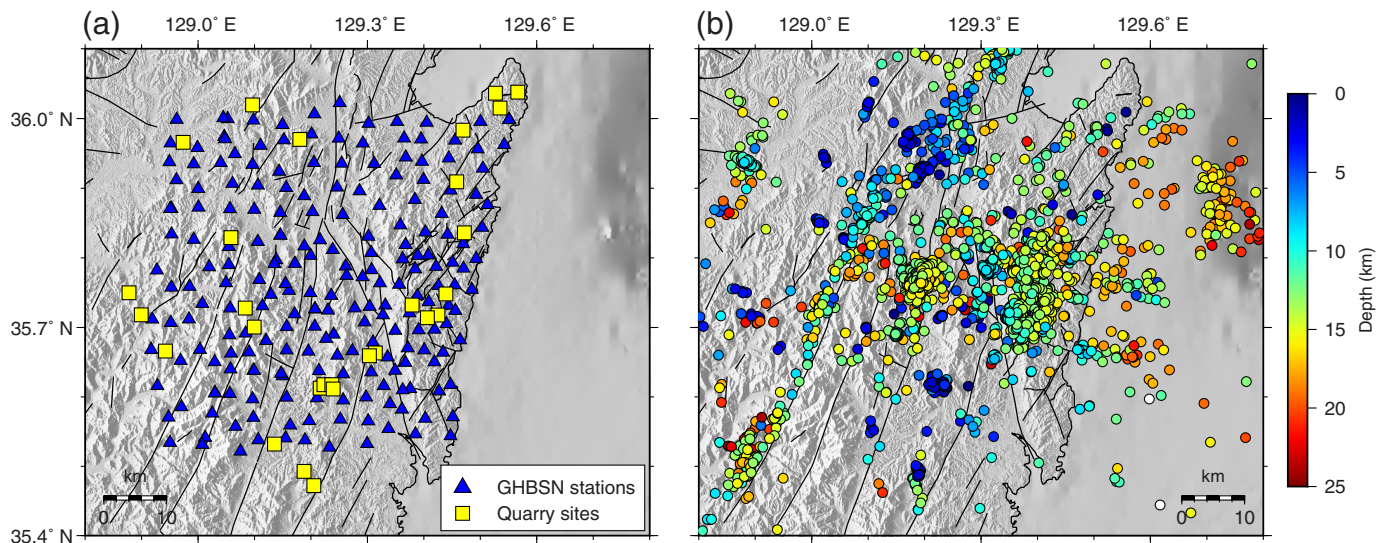


Figure 1. Geological–physiographic map of the study area showing fault lineaments (black lines; modified following [Kee et al., 2019](#)) and two orientations of the maximum horizontal stress. The red arrows indicate regional S_{Hmax} orientation compiled from [Chang et al. \(2010\)](#) and [Soh et al. \(2018\)](#). The blue arrows are local S_{Hmax} orientations east of the Ulsan fault (USF) inferred by [Seo \(2021\)](#); blue box region). Major structures are labeled:

DRF, Dongrae fault; IGF, Ilgwang fault; MiRF, Miryang fault; MoRF, Moryang fault; USF, Ulsan fault; YSF, Yangsan fault; and YTL, Yeonil tectonic line. The red star marks the epicenter of the 2016 M_L 5.8 Gyeongju earthquake. The inset figure is the regional map of northeast Asia, including the Korean Peninsula (KP), highlighting the study area (red box). The color version of this figure is available only in the electronic edition.



The network expanded from seven to 200 stations between November 2017 and September 2019, covering an area of $60 \times 60 \text{ km}^2$ (Fig. 2; Heo *et al.*, 2024). The GHBSN was originally deployed as a dense post-earthquake monitoring network. Although the present study uses data recorded between January 2019 and December 2020, the network continued to operate beyond this period: 200 stations were maintained until 2024, and about 150 stations remain operational following partial redeployment and southward expansion. Initial event detections were performed automatically, and the events were manually relocated using the Hypoellipse algorithm (Lahr, 1999) and the 1D regional velocity model for the GB (Kim *et al.*, 2011), following the procedure of Heo *et al.* (2024). Between January 2019 and December 2020, the GHBSN recorded 3159 local seismic events having depths less than 25 km. Relocated seismicity was concentrated near the 2016 Gyeongju earthquake region and in the northeastern hanging wall of the USF, which is an NNW-striking and ENE-dipping reverse fault. Quarry-related events were identified from waveform characteristics and hypocentral distributions during manual relocations and were excluded from the interpreted dataset. These events typically clustered near known quarry sites, had shallow depths ($<5 \text{ km}$), and showed unclear *P*- and *S*-phase onsets.

SWS measurements

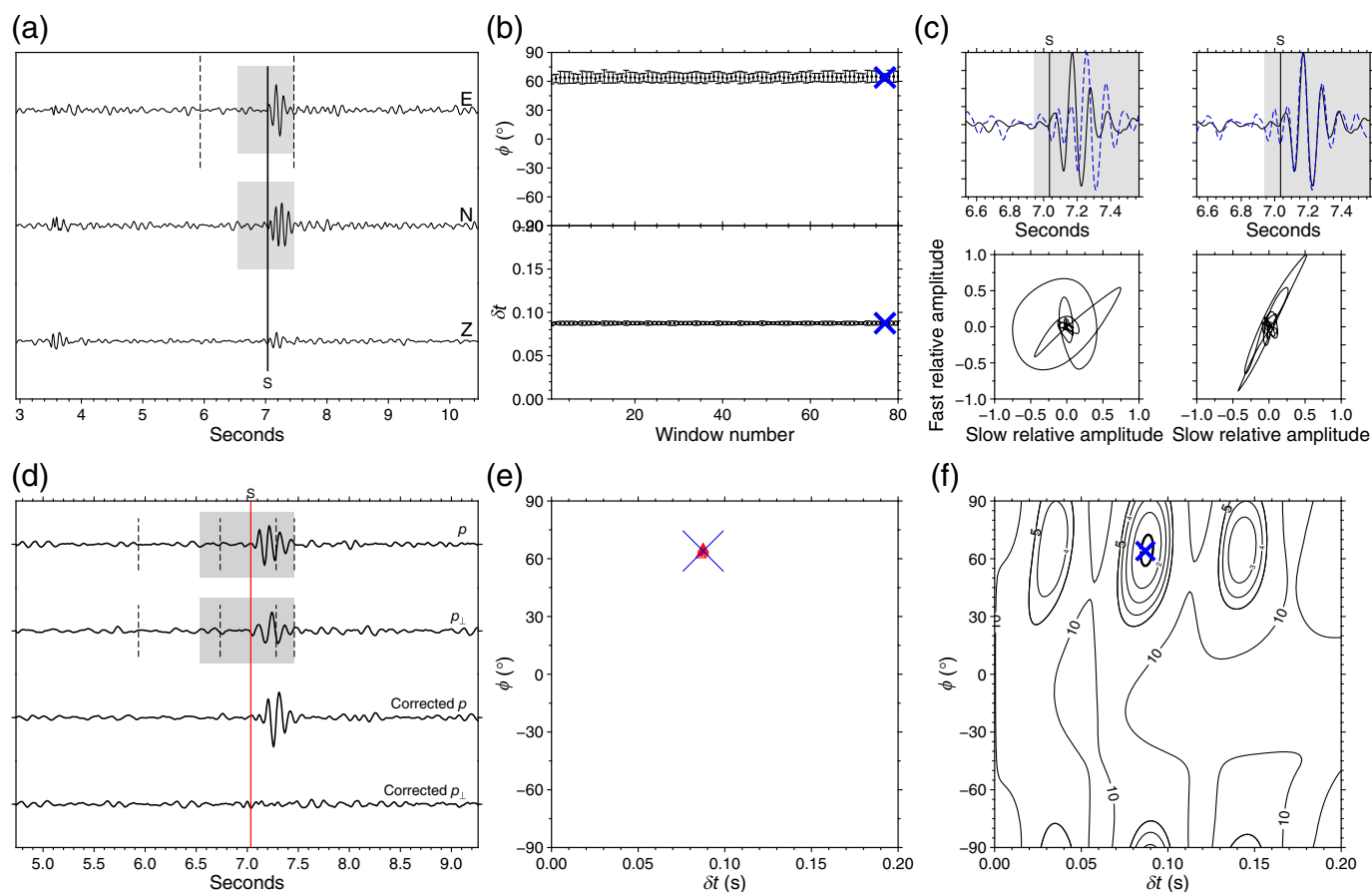
We performed the splitting analysis on the two horizontal components, east (E) and north (N), without applying an LQT rotation. This choice allowed the measured fast directions to be referenced directly to geographic coordinates. Potential contamination from phases outside the shear-wave window was reduced by applying the distance–depth criterion and by visually inspecting the candidate measurements. The MFAST program (Savage *et al.*, 2010) uses the eigenvalue method (Silver and Chan, 1991; Walsh *et al.*, 2013) to estimate the two SWS parameters, fast direction (ϕ) and delay time (δt). It also incorporates cluster analysis of SWS parameters

Figure 2. Station and seismicity maps of the Gyeongju Hi-density Broadband Seismic Network (GHBSN). (a) GHBSN stations deployed in response to the 2016 M_w 5.5 (M_L 5.8) Gyeongju earthquake are shown by blue triangles, and known quarry sites are shown by yellow squares. (b) Local earthquakes used in this study are color coded by hypocentral depth. Quarry blasts identified during manual relocation were excluded from the interpreted earthquake catalog. The color version of this figure is available only in the electronic edition.

following Teanby *et al.* (2004) (Fig. 3e). A sliding-window approach (Fig. 3b) was employed to improve the robustness of parameter selection and provide error ranges for ϕ and δt .

In addition, MFAST applies multiple band-pass filters to enhance the signal-to-noise ratio (SNR) and ranks the resulting candidate solutions. Filtering improves waveform clarity without biasing the final SWS parameters (Savage *et al.*, 2010). Multiple band-pass filters were applied to each waveform, yielding candidate splitting solutions under different filtering conditions (Table 1). For subsequent quality control, we retained up to three best-ranked filter-specific candidate measurements (filter bands 1–3, fb1–fb3) for each event–station waveform. Across the analyzed local-earthquake dataset before quality screening, this procedure yielded 160,861 filter-specific candidate measurements. Unless otherwise noted, all counts reported below refer to filter-specific candidate measurements rather than unique event–station waveforms.

Null measurements were also obtained. These occur when the difference between the initial shear-wave polarization and the analyzed fast polarization falls within a prescribed interval (Peng and Ben-Zion, 2004). Although null measurements do not provide reliable delay-time estimates, they may, in simple anisotropic media, provide a quantitative check on fast-polarization azimuths because the initial shear-wave polarization is expected to be parallel or perpendicular to the fast axis. We did not use null measurements quantitatively here because path-integrated splitting through heterogeneous anisotropic domains can make such interpretations ambiguous.



Data selection criteria

The candidate measurements were then screened using three criteria. First, to avoid contamination by phases outside the shear-wave window, we required the epicentral distance to be less than 1.5 times the hypocentral depth of the event. Second, we retained only measurements with fast-direction (ϕ) uncertainties $<15^\circ$. Third, null results were excluded, defined as cases in which the angular difference between the analyzed fast polarization and the initial shear-wave polarization was $\leq 20^\circ$. Applied individually, the distance–depth criterion retained 96,474 candidate measurements, the ϕ -uncertainty criterion retained 141,672, and the nonnull criterion retained 142,669; application of all three criteria retained 73,313 candidate measurements.

We did not apply a strict uncertainty cutoff to delay time (δt) because δt estimates are strongly affected by cycle skipping. Because automatic procedures cannot fully eliminate cycle-skipped or internally inconsistent solutions, we manually compared the fb1–fb3 candidate measurements for each event–station waveform and retained the best measurement where a reliable solution could be identified. Among mutually consistent candidates, we preferentially selected the solution with the smallest ϕ and δt uncertainties. This procedure yielded a final catalog of 4611 reliable SWS measurements from 1602 local earthquakes, which were used in the subsequent analyses.

Figure 3. Example output of the automatic shear-wave-splitting (SWS) analysis. Panels (from top-left to bottom-right) show: (a) filtered ENZ waveforms; (b) window-by-window estimates of fast direction (ϕ) and delay time (δt); (c) waveforms along the fast and slow axes and the corresponding horizontal particle motion; (d) uncorrected and SWS-corrected horizontal waveforms in the rotated coordinate frame; (e) scatter plot of δt versus ϕ ; and (f) F -test-based misfit contour map of the SWS analysis. The color version of this figure is available only in the electronic edition.

TABLE 1

Frequency Bands Used in This Study and the Number of Measurements in Each Band

Lower Cutoff Frequency (Hz)	Upper Cutoff Frequency (Hz)	Number of Measurements
1	15	1375
1	30	1159
1	8	449
2	8	4
3	15	8
3	30	1159
5	10	1
5	30	21
5	45	419
10	45	16

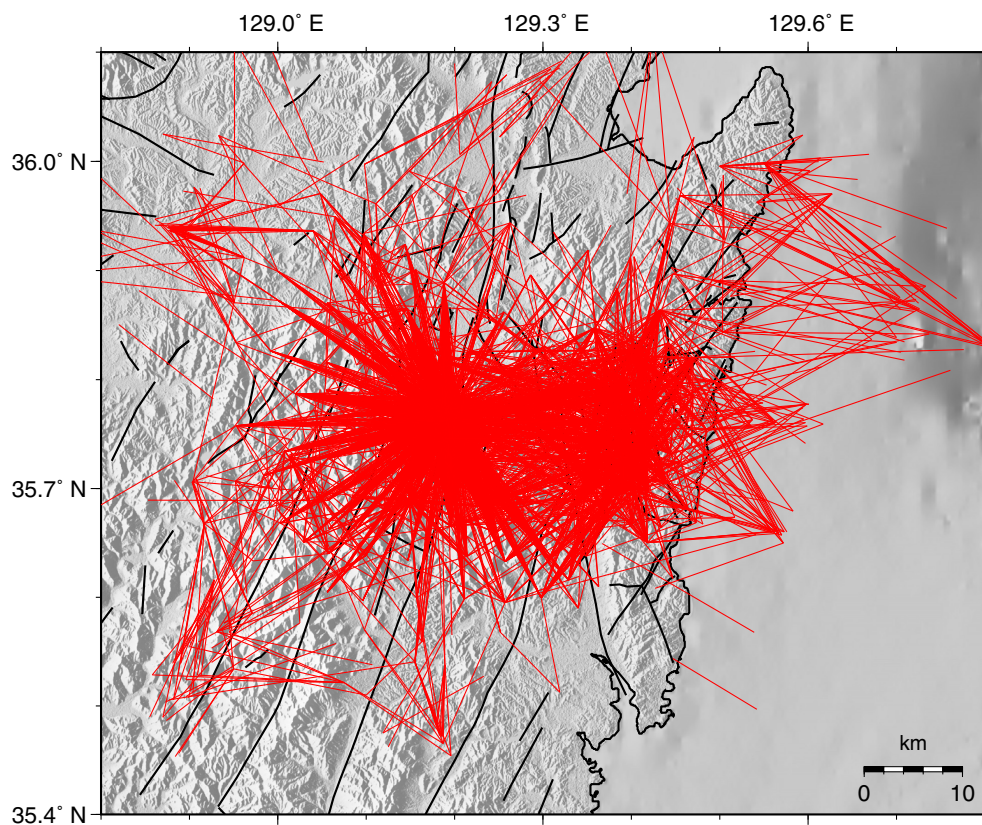


Figure 4. Ray path coverage (red lines) for measurements yielding reliable SWS parameters. The color version of this figure is available only in the electronic edition.

Station orientation correction

For the principal component analysis (PCA) estimates of [Seo et al. \(2022\)](#), most station–segment estimates (238 of 335 time-varying segments across 200 stations) showed misorientation angles of less than 15° from true north, whereas three stations were misoriented by more than 150° (N131, N160, and N199). We corrected the measured fast directions by subtracting the station-specific mean misorientation angle estimated from the PCA method. For the two-horizontal-component eigenvalue method used here, this postanalysis correction is equivalent to prerotating the horizontal waveforms because a constant rotation of the horizontal coordinate frame only changes the reference azimuth of the recovered fast direction, whereas the delay time and the eigenvalue-based linearity criterion remain unchanged.

Delay-time tomography and spatial averaging of anisotropy

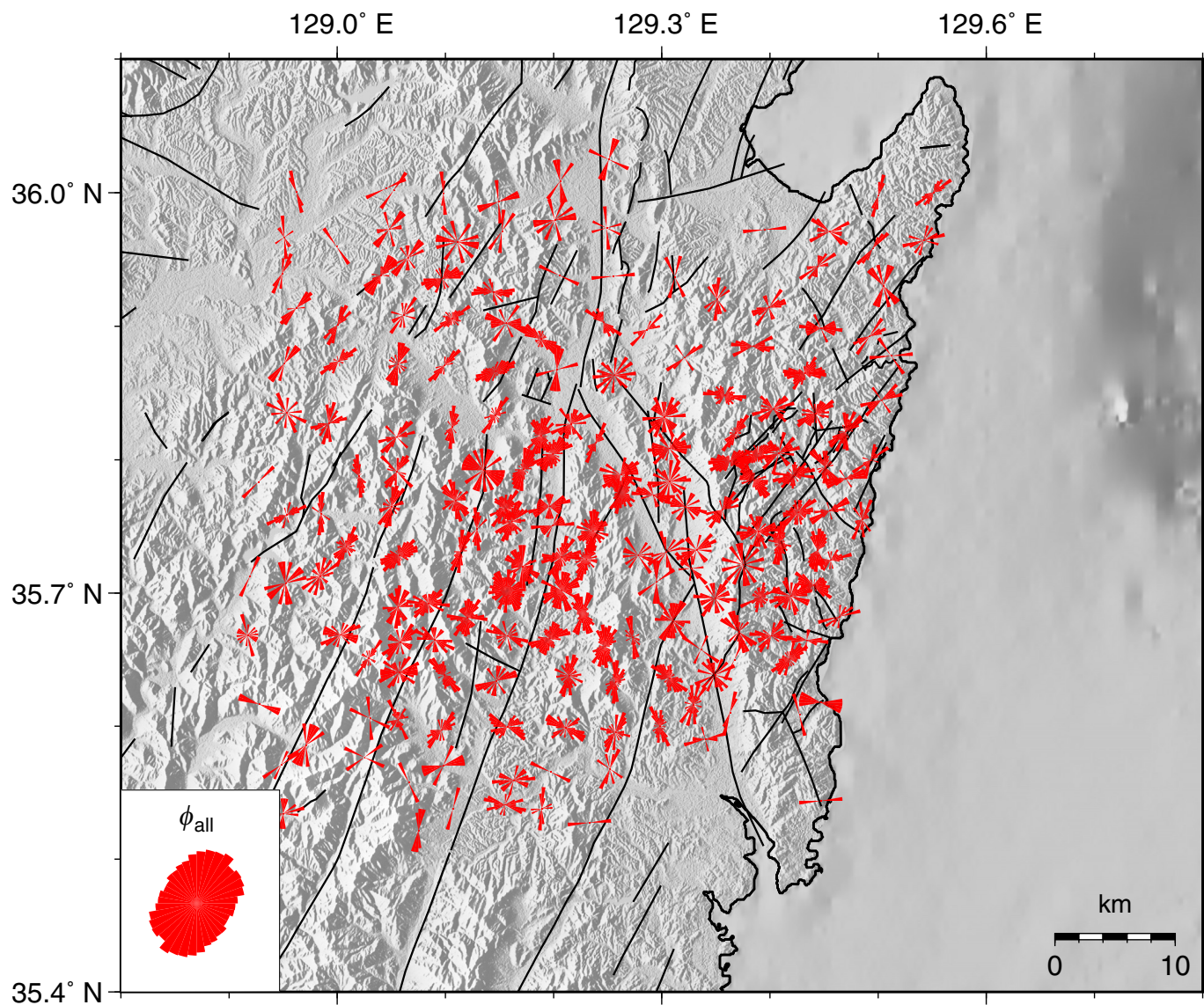
We analyzed the GHBSN SWS dataset using the TESSA framework of [Johnson et al. \(2011\)](#), with analysis parameters adapted to the GHBSN dataset. Delay-time tomography was used to estimate the first-order lateral distribution of relative anisotropy strength from path-integrated delay times (δt), under the simplifying assumption that δt is additive along each ray path ([Johnson et al., 2011](#)). Spatial averaging was used to identify representative

fast-direction patterns on the same regular 2-km grid, with a minimum of 15 ray paths per cell for both analyses. For spatial averaging, fast directions were weighted by the inverse square of the distance from the station, following the $1/d^2$ scheme adopted by [Johnson et al. \(2011\)](#). Because the ϕ distributions in our dataset were commonly bimodal or multimodal, representative orientations were identified from local maxima rather than from circular-mean directions based on the von Mises framework ([Mardia, 1975](#)). To ensure robust peak selection, neighboring local maxima were required to differ by more than 30° , and each selected maximum was required to exceed 10% of the total weight; this approach is conceptually similar to the peak-separation and threshold criteria used by [Ortega-Romo et al. \(2021\)](#).

RESULTS

We obtained 4611 reliable SWS measurements from 1602 local earthquakes after automatic screening of 160,861 filter-specific candidate measurements and subsequent manual review. Several independent diagnostic tests confirmed that the dataset was of high quality (Fig. 3). Fast and slow shear waves were clearly separated for representative events, and the corrected particle motions exhibited nearly linear trajectories, indicating that the splitting correction effectively eliminated anisotropic effects (Fig. 3c). Moreover, after correction, one of the horizontal components exhibited amplitudes close to the background noise level (Fig. 3d), further supporting the robustness of this procedure. When multiple filter bands were applied, both ϕ and δt remained consistent, demonstrating that the results were not artifacts of the chosen frequency band but reflected stable anisotropic signatures. These quality checks ensure that subsequent interpretations are based on a reliable, well-resolved dataset. The main ranges of the applied filters are (1, 15), (1, 30), and (3, 30) in Hertz, accounting for 1375, 1159, and 1159 measurements, respectively, or approximately 80% of the final catalog (Table 1).

The spatial coverage of the SWS measurements (Fig. 4) was densest around the epicentral region of the 2016 Gyeongju earthquake sequence and in the USF hanging wall (eastern block). This distribution reflects the concentration of seismic



activity in the southeastern Korean Peninsula during the observation period. In contrast, fewer ray paths were available in the northern, northeastern, and southwestern portions of the Gyeongju earthquake sequence and in the southern part of the USF hanging wall, resulting in coverage gaps that should be considered when interpreting spatial averages and tomographic results. Nevertheless, the dense coverage around the Gyeongju earthquake sequence and the USF enabled a detailed investigation of upper-crustal anisotropy in an intraplate setting.

At the individual-station level (Fig. 5), ϕ and δt showed substantial spatial variability, and no simple large-scale crustal pattern was evident before spatial averaging. Stations near the YSF commonly showed N–S to NNE–SSW fast orientations, whereas stations near the USF showed more scattered orientations.

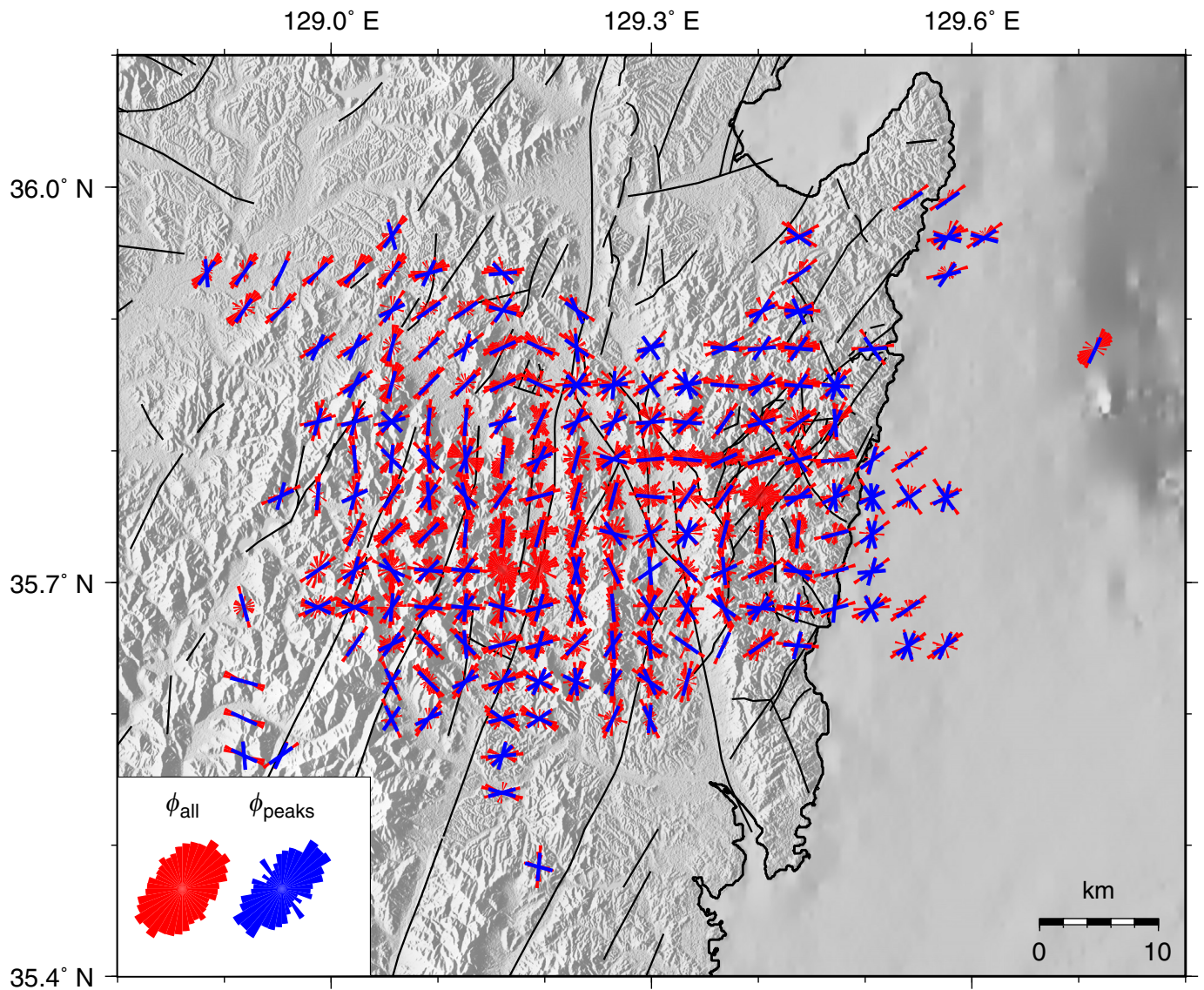
After spatial averaging, a dominant NE–SW fast orientation emerged across most of the study area (Fig. 6). Local deviations were also evident. Around the YSF, the average ϕ rotated to N–S and NNE–SSW. Near the USF, ϕ shifted to E–W, and east of the USF, NE–SW and E–W orientations coexisted. Delay-time

Figure 5. Spatial distribution of station-based rose diagrams showing fast polarization directions; the bottom-left inset displays the aggregate fast-polarization directions. The color version of this figure is available only in the electronic edition.

tomography (Fig. 7) showed elevated δt values ($> \sim 0.005$ s/km) in the epicentral area of the 2016 Gyeongju earthquake sequence, east of the USF, and along two linear anomalies north of the Gyeongju earthquake sequence. In contrast, west of the USF, δt values were generally lower, with only small isolated patches of elevated anisotropy. A noteworthy feature is that the high-anisotropy zone east of the USF terminates along the YTL, rather than along the USF. The maximum observed δt was ~ 0.2 s.

DISCUSSION

The maximum observed δt (~ 0.2 s) is comparable with crustal splitting values reported from other stable continental settings, including the Central North China craton and southern



African cratons (Thybo *et al.*, 2019; Zheng *et al.*, 2019). The main significance of the present dataset, therefore, lies not in an unusually large anisotropy magnitude, but in its strong spatial concentration around specific fault systems and tectonic domains.

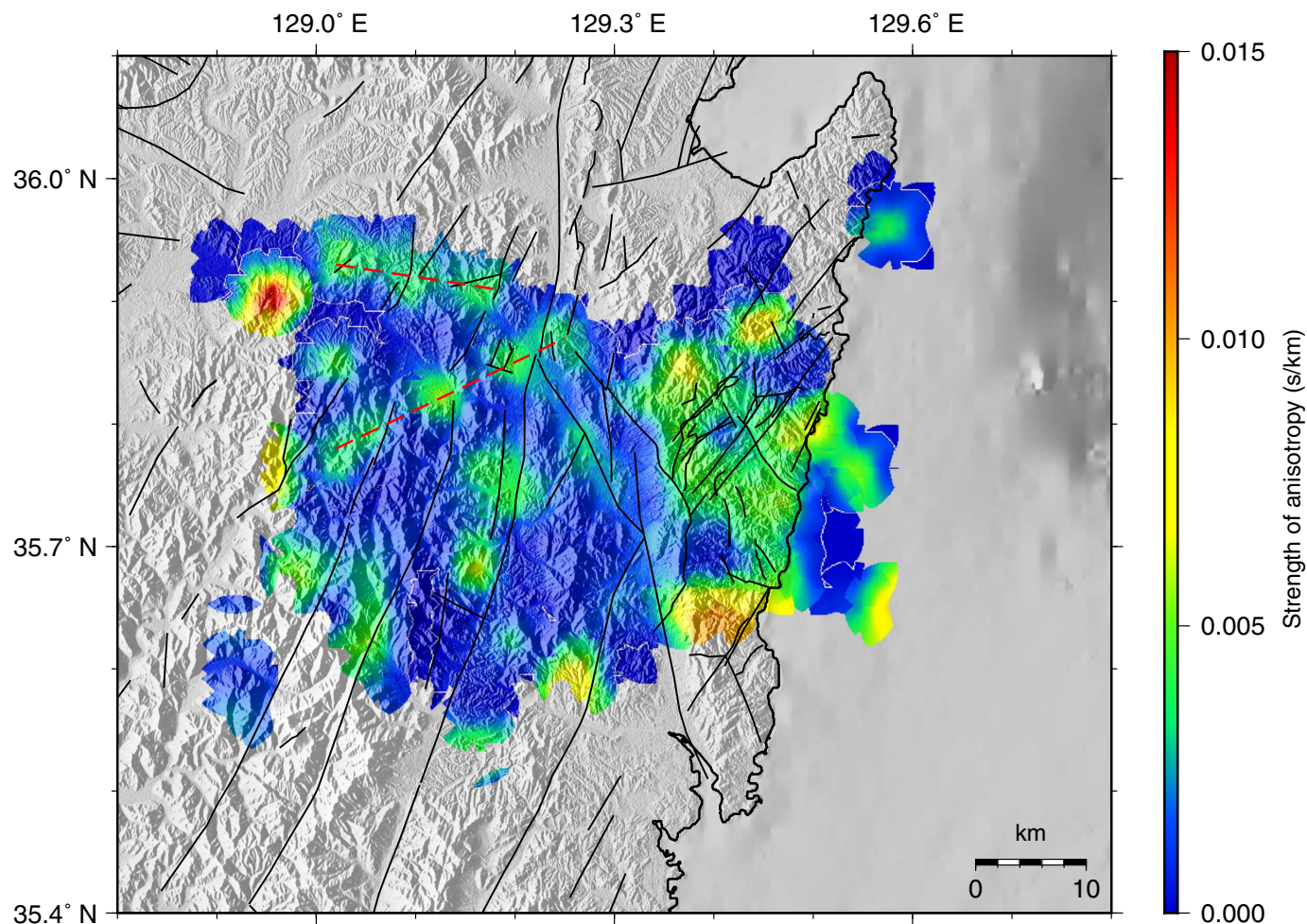
Structural controls on anisotropy

The SWS results indicate that upper-crustal anisotropy in the study area is spatially heterogeneous and reflects multiple anisotropic sources. Faults and fractures are widely recognized as major contributors to seismic anisotropy (Crampin and Lovell, 1991; Zinke and Zoback, 2000; Kaviris *et al.*, 2018). Along the YSF, MiRF, and MoRF, many fast polarization directions are sub-parallel to the mapped fault strikes, suggesting that these fault systems and associated fracture fabrics provide an important structural control on anisotropy. The YSF is a mature and long-lived strike-slip fault system in the southeastern Korean Peninsula (Okada *et al.*, 1994; Cheon *et al.*, 2019; Kim *et al.*, 2021), and its prolonged deformation history likely contributed to the development of fault-related anisotropic fabrics.

Figure 6. Spatial distribution of fast-direction rose diagrams (red) and their local maxima (blue) for shear-wave splitting. Fast directions were spatially averaged using inverse-distance-square weighting and gridded at 2 km resolution. The color version of this figure is available only in the electronic edition.

However, the strongest anisotropy is not distributed continuously along the entire YSF. Instead, delay-time tomography shows that elevated anisotropy is localized near the 2016 Gyeongju earthquake sequence, particularly around the Naenam fault, the rupture zone of the sequence (Fig. 7; Woo *et al.*, 2019; Heo *et al.*, 2024). This spatial concentration suggests that recent seismic deformation associated with the Gyeongju sequence contributed to the development or maintenance of anisotropic fabrics in this zone, whereas the main YSF trace outside the sequence does not show equally strong anisotropy.

Two additional linear high- δt features north of the Gyeongju earthquake sequence are also visible in Fig. 7



(latitude $> 35.7^\circ$ and longitude $< 129.3^\circ$; ESE–WNW and NE–SW trends). Because these features are not independently linked here to mapped structures or concentrated seismicity, and because spatial resolution decreases outside the best-sampled areas, their tectonic significance remains uncertain. They may reflect localized fracture zones or unmapped structures, but we interpret them cautiously in the present study.

Anisotropic domain east of the Ulsan fault and the role of the Yeonil tectonic line

In the eastern part of the study area, anisotropy changes sharply across the USF. West of the USF, representative ϕ s are mainly N–S, whereas east of the fault they shift to NE–SW and E–W. Delay-time tomography further shows that elevated δt is concentrated east of the USF and that the high-anisotropy zone terminates along the YTL rather than along the USF trace. These observations indicate that the USF marks a transition between anisotropic domains, whereas the YTL defines the eastern boundary of the strongly anisotropic zone.

The YTL and the faults east of the USF are associated with Miocene basin development and NE-trending basin-bounding fault systems (Kim *et al.*, 1998; Son *et al.*, 2015; Cheon, Shin, *et al.*, 2023). These basin-related structures may have enhanced fracture density and locally promoted crack opening during

Figure 7. Spatial distribution of delay-time strength (s/km) derived using the TESSA procedure of Johnson *et al.* (2011), gridded at 2-km resolution. Two linear features mentioned in the text are marked with red dashed lines. The color version of this figure is available only in the electronic edition.

extensional-to-transensional deformation associated with basin formation. Thus, the elevated anisotropy east of the USF may partly reflect open or preferentially aligned fracture fabrics inherited from Miocene basin-bounding deformation. These inherited structures likely contribute to the NE–SW anisotropy of the fabric. Faulting styles also differ across the YTL, with reverse faulting to the west and strike-slip faulting to the east (Cheon, Kim *et al.*, 2023), indicating that the YTL represents a major structural boundary. The anisotropy east of the USF, therefore, likely reflects a combination of inherited structural fabrics and the present regional stress field, rather than a single mechanism.

Stress regime and depth-dependent anisotropy

The southeastern Korean Peninsula has been under ENE–WSW compression since the Quaternary (Cheon *et al.*, 2019). In the southern part of the study area, and locally east of the USF, E–W to ENE–WSW fast orientations are broadly

consistent with this regional stress field (Soh *et al.*, 2018; Kim and Chang, 2021; Seo, 2021). Such orientations can reflect stress-induced crack alignment under the present compressive stress regime (Boness and Zoback, 2006; Syuhada *et al.*, 2016).

In contrast, NE–SW polarization dominates the western YSF, even in areas between the YSF-parallel faults. These orientations were not aligned with the mapped faults or the regional ENE–WSW stress regime. Yale (2003) noted that the anisotropy inferred from the SWS can also be influenced by shallow stress fields that differ from those in the deeper regime. Ree *et al.* (2021) suggested that the compressive stress orientation may vary with depth, documenting subhorizontal E–W compression at shallow depths (3–6 km) in Pohang, located just north of the study area. Similarly, Kang *et al.* (2023) reported NE–SW and E–W compressive stress regimes in the area immediately south of the study region (Ulsan and Busan). Considering the two shared geological features, the tectonic province (GB) and the YSF fault system, similar shallow stress patterns were expected throughout the study area. However, at greater depths (> 6 km), the stress regime is more likely to be ENE–WSW (Ree *et al.*, 2021; Kang *et al.*, 2023). This depth-dependent stress field may indicate that part of the NE–SW anisotropy originates in a thin upper-crustal layer influenced by shallow stress conditions, whereas deeper anisotropy may reflect the broader ENE–WSW regional compression.

To evaluate the depth dependence of the observed anisotropy, we examined δt and ϕ as functions of hypocentral depth for the final SWS catalog (Fig. S1). The δt values do not show a systematic increase with event depth, suggesting that the observed splitting is unlikely to be dominated by a vertically pervasive anisotropic volume that accumulates delay time monotonically with path length. Instead, the data are more consistent with laterally localized anisotropic domains, including fault-related and basin-related fracture fabrics. We also considered whether topographic stresses could contribute to the observed anisotropy. However, the strongest anisotropy occurs mainly near the Gyeongju earthquake sequence and east of the USF, rather than being systematically organized by coastline-parallel or topographic gradients. We therefore regard topographic stress as a possible local effect, but not as a first-order control on the anisotropy pattern resolved in this study.

Seismicity and slip tendency

Seismicity provides an independent indicator of ongoing crustal deformation. In the southeastern Korean Peninsula, seismic activity was primarily concentrated in the 2016 Gyeongju earthquake sequence and east of the USF (Fig. 2). The Gyeongju sequence occurred at depths of 12–16 km (Woo *et al.*, 2019; Heo *et al.*, 2024), which overlap the zones of high δt in the delay-time tomography (Fig. 7). This spatial association suggests that the SWS results are sampling anisotropies within the seismogenic crust affected by the recent earthquake sequence. A

similar association between elevated δt and active seismicity is observed east of the USF, suggesting that long delay times mark crustal volumes that have undergone stronger deformation or both contain more developed crack or fracture fabrics.

Slip tendency is commonly quantified as the ratio of resolved shear stress to effective normal stress on a fault plane under the assumed stress field, with higher values indicating a greater likelihood of reactivation. Delay-time tomography also shows a broad spatial correspondence with the slip-tendency pattern of major faults between the YSF and the Ulleung fault, an offshore fault along the eastern Korean coastline, as reported by Kim *et al.* (2023). Faults east of the USF have high slip-tendency values, whereas the YSF and USF show lower values, consistent with the concentration of elevated δt east of the USF. In the Gyeongju area, however, the comparison is less direct because the slip-tendency analysis applies to the YSF, whereas the strongest anisotropy is localized near the Naenam fault. Slip tendency, therefore, provides useful regional context for reactivation potential, but it should not be interpreted as a one-to-one predictor of seismic anisotropy.

Lithological influences

Lithology may also modulate the anisotropy pattern, although the evidence is less direct than for structural controls (Teufel, 1989). The study area includes granitic rocks and Cretaceous and Cenozoic sedimentary units (Fig. S2; Chough *et al.*, 2000; Kim *et al.*, 2023). No clear anisotropic contrast is observed between the western and eastern YSF, suggesting that the limited granitic exposures along the western YSF are not a first-order control on the SWS pattern. In contrast, east of the USF, where Cenozoic basins and basin-bounding faults are developed, both ϕ and δt differ from those west of the fault. This contrast may partly reflect lithologic variation across the Cretaceous–Cenozoic transition, but it is more likely to reflect the deeper fault architecture associated with the Miocene basins.

Strong anisotropy zones overlap several basins east of the USF, except in the Janggi and Jeongja basins, where data coverage is limited. Because basin fills themselves are unlikely to generate coherent anisotropy, a more plausible source is the NE-trending basin-bounding faults, which extend to greater depths and can host deformation (Son *et al.*, 2015). Lithologic effects, therefore, may be secondary, whereas basin-related structures are likely more important. Because the subsurface distribution of rock units is not directly resolved here, lithologic interpretations remain tentative.

Additive assumption of anisotropic parameters

The additive assumption is appropriate as a first-order approximation when the ray-path samples anisotropic domains with broadly similar fast-axis orientations or when the resulting delay times are interpreted as relative measures of anisotropy strength rather than as a unique depth-resolved anisotropic model. The assumption can break down if a ray path crosses multiple

anisotropic layers or domains with strongly different fast directions because splitting from different segments may interfere nonlinearly or partially cancel (Hudson *et al.*, 2023). In the present dataset, the local-earthquake ray paths are relatively short and densely sample the Gyeongju sequence and the eastern USF domain (Fig. 4), and the averaged fast-direction map shows coherent domain-scale patterns (Fig. 6). We therefore use TESSA delay-time tomography (Fig. 7) to identify first-order lateral variations in relative anisotropy strength, while avoiding a unique interpretation of the results as a fully depth-resolved anisotropic structure.

LIMITATIONS

Several limitations should be considered when interpreting these results. First, SWS measurements are path-integrated and may average multiple anisotropic sources along each ray path. Second, event distribution is uneven, with dense coverage near the Gyeongju sequence and east of the USF, but sparse coverage elsewhere, limiting resolution in poorly sampled areas (Fig. 4). Third, delay-time tomography depends on the spatial and depth distribution of earthquakes. Fourth, lithological interpretations are indirect because subsurface distributions are inferred mainly from surface geology. Finally, TESSA-based averaging (Johnson *et al.*, 2011) assumes homogeneous anisotropy within each grid cell, which may oversimplify local heterogeneity. Despite these limitations, consistency among the averaged ϕ map, delay-time tomography, and known tectonic structures supports the interpretation that crustal anisotropy in the southeastern Korean Peninsula reflects both inherited structures and present-day deformation.

CONCLUSIONS

We obtained 4611 reliable SWS measurements from 1602 local earthquakes recorded by the GHBSN to investigate upper-crustal anisotropy in the southeastern Korean Peninsula. Spatial averaging revealed a dominant NE–SW fast-polarization orientation across much of the study area, with local rotations to N–S and NNE–SSW around the YSF and to E–W near the USF; east of the USF, NE–SW and E–W orientations coexist. Elevated δt values are concentrated near the 2016 Gyeongju earthquake sequence and east of the USF, and the eastern high-anisotropy domain terminates along the YTL rather than along the USF trace.

These observations indicate that upper-crustal anisotropy in the study area is controlled by multiple mechanisms rather than a single source (Graham *et al.*, 2020). Along the YSF system, fault-related fabrics are important, and the strongest anisotropy is localized near the Naenam fault and the Gyeongju sequence. East of the USF, anisotropy likely reflects a combination of inherited Miocene basin-bounding structures and the present regional stress field. The widespread NE–SW component, therefore, should not be attributed to a single origin; instead, it likely records domain-dependent contributions from inherited structures and, locally, shallow stress conditions distinct from the

deeper ENE–WSW regional compression. This study provides a baseline for future work on upper-crustal anisotropy and intra-plate fault-zone deformation in southeastern Korea.

DATA AND RESOURCES

Waveform data from the Gyeongju Hi-density Broadband Seismic Network are managed by the Korea Institute of Nuclear Safety (<https://www.kins.re.kr>, last accessed July 2026) and are available upon reasonable request. Shear-wave splitting (SWS) was performed using the MFAST package (code and user's manual available at <https://mfast-package.geo.vuw.ac.nz>; last accessed March 2026; the multiple band-pass filters are listed in Table 1). The TESSA package was used for delay-time tomography and spatial averaging of SWS parameters and is available via <https://sites.google.com/site/jessicahelenjohnson/tessa> (last accessed March 2026). The Hypoellipse program for the hypocenter determination of the events used in this study can be accessed via <https://pubs.usgs.gov/of/1999/ofr-99-0023/> (last accessed March 2026). Phase arrivals were determined using the Seismic Analysis Code program (Goldstein *et al.*, 2003; Goldstein and Snoke, 2005). Figures were generated using Generic Mapping Tools version 5 (Wessel *et al.*, 2013). The final catalog of 4611 reliable SWS measurements and the corresponding MFAST summary files are archived in Zenodo at doi: [10.5281/zenodo.20564961](https://doi.org/10.5281/zenodo.20564961). The catalog includes event and station information, selected filter bands, ϕ , δt , and associated uncertainties. The supplemental material for this article includes the scatter plots of SWS parameters as functions of hypocentral depth, and the map of first-order lithological units.

DECLARATION OF COMPETING INTERESTS

The authors acknowledge that there are no conflicts of interest recorded.

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